Chapter 7: Behavioral Strategies

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

Updated 2026-11-04 at 23:02:51

Where We Left Off

Chapter 4: game trees with **information sets**, solved by SPNE

- Solve the smallest subgames first, work backwards
  - A generalization of backwards induction
- We said if a subgame has **no** NE, then there's **no SPNE**

Chapter 6: **mixed strategies**

- **Nash's Theorem**: every finite game (or subgame) has a (possibly mixed) NE
- So the "no NE in a subgame" problem can be fixed up
- But we have to allow mixing inside a game tree

Chapter 7:

1. Put lotteries and mixing into game trees
2. Find a **simpler** way to mix in trees: behavioral strategies
3. Show that a SPNE always exists
4. See what SPNE still gets wrong

Lotteries in Game Trees

“The Game” 2013

2013 OSU @ Michigan. Urban Meyer vs. Brady Hoke. 1,129 total offensive yards
Was 42-35. With 32 sec left, MI scores TD. Now 42-41.

- Brady Hoke: **Kick** to tie/OT or **Go for 2** to win

- **Kick**: Good 95% of the time, almost surely leads to overtime
  - Overtime: let’s say it’s 50-50 who wins
- **Go for 2**: Model it as a 2×2 subgame
  - Michigan: **Run** or **Pass**
  - OSU: **Stop Run (SR)** or **Stop Pass (SP)**

Outcome Payoffs: $u(\text{Win}) = 100$, $u(\text{Lose}) = 0$ for both teams

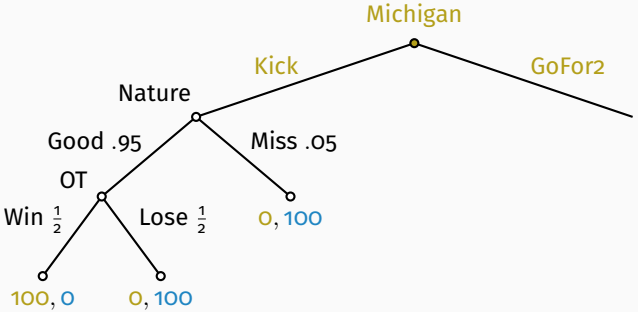
How to draw this tree?

Drawing the Tree



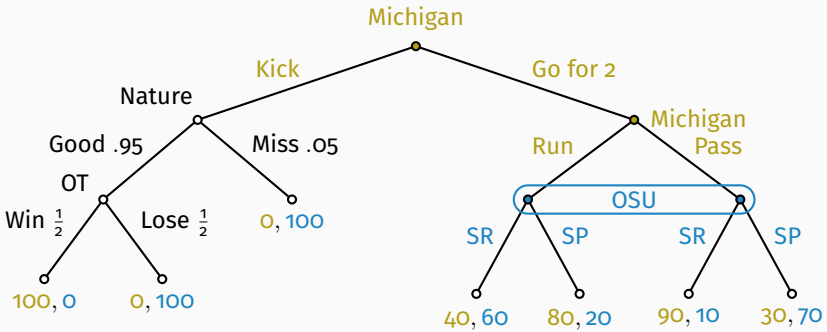
Michigan moves first

Drawing the Tree



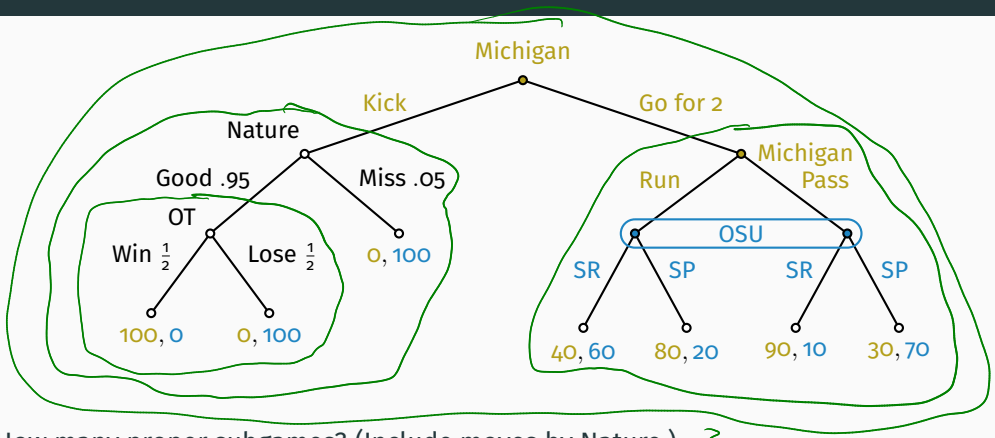
Kick: two moves by “Nature”. Each branch has a probability

Drawing the Tree



GoFor2: a simultaneous-move game hanging off the tree

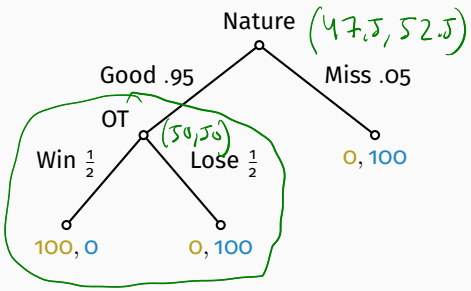
Drawing the Tree



How many proper subgames? (Include moves by Nature.) 3

Nature Gives a Lottery Over Outcomes

After Michigan chooses Kick:



Overtime gives Michigan $\begin{pmatrix} \text{Win} & \text{Lose} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$U(\text{OT}) = \frac{1}{2}(100) + \frac{1}{2}(0) = 50$ OSU : 50

Kick gives Michigan $\begin{pmatrix} \text{OT} & \text{Lose} \\ .95 & .05 \end{pmatrix}$

$U(\text{Kick}) = 0.95(50) + 0.05(0) = 47.5$

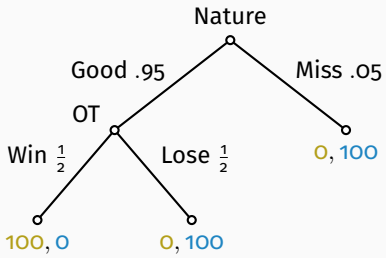
OSU : 52.5

Rule: Same as Chapter 6. Use expected utilities

Bonanno (§7.1) writes the lottery as the payoff instead of drawing Nature. Using Nature is more standard.

Nature Gives a Lottery Over Outcomes

After Michigan chooses Kick:



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$U(\text{OT}) =$ _____

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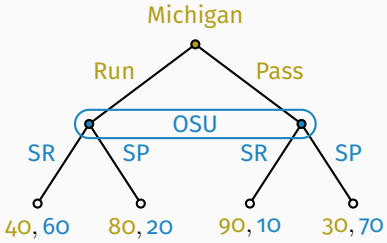
Rule: Same as Chapter 6. Use expected utilities

So: replace the entire Kick branch with (47.5, 52.5)

Bonanno (§7.1) writes the lottery as the payoff instead of drawing Nature. Using Nature is more standard.

Subgame-Perfect Equilibrium: Allowing Mixing

Step 1: Solve the Subgame After "Go for 2"



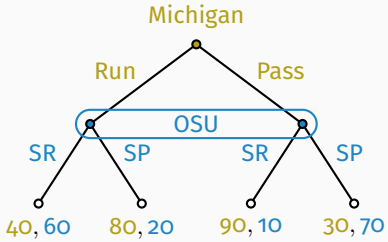
OSU

Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

No pure NE

Payoffs = Expected utility = Prob. of Win

Step 1: Solve the Subgame After "Go for 2"

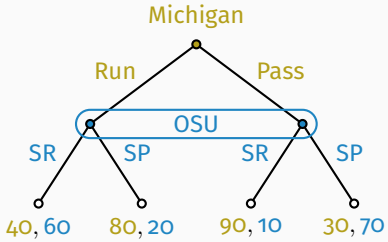


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Chapter 4: "there's no SPNE" Chapter 6: We can find MSNE...

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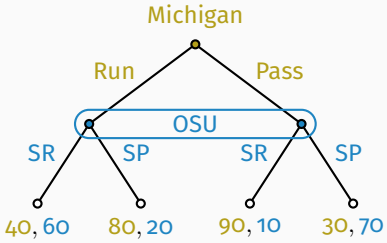


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Chapter 7: let's include subgame MSNE in our definition of SPNE.

Step 1: Solve the Subgame After "Go for 2"



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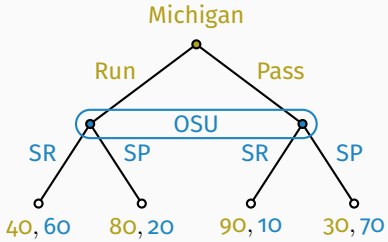
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Michigan's p on Run that makes OSU indifferent:

$U(SR) = 60p + 10(1-p) \quad U(SP) = 20p + 70(1-p) \quad U(SR) = U(SP) \Rightarrow p^* = 3/5$

Step 1: Solve the Subgame After "Go for 2"

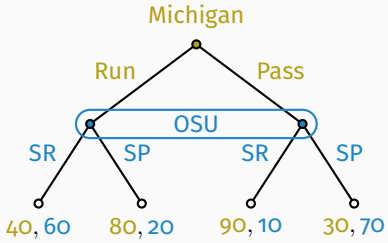


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OSU's q on SR that makes Michigan indifferent:
$U(Run) = 40q + 80(1-q)$ $U(Pass) = 90q + 30(1-q)$ $U(Run) = U(Pass) \Rightarrow q^* = 1/2.$

Step 1: Solve the Subgame After "Go for 2"



	OSU: Stop Run	OSU: Stop Pass
Michigan: Run	40, 60	80, 20
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Payoffs = Expected utility = Prob. of Win

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OSU's q on SR that makes Michigan indifferent:

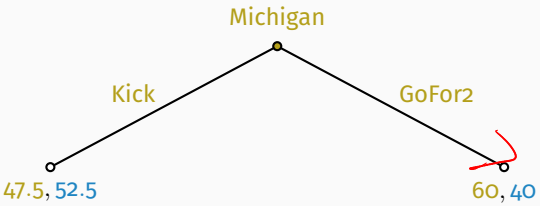
$U(Run) = 40q + 80(1-q) \quad U(Pass) = 90q + 30(1-q) \quad U(Run) = U(Pass) \Rightarrow q^* = 1/2.$

Expected Payoffs in MSNE:

Michigan: 60 (percent chance of winning)

OSU: 40 (percent chance of winning)

Step 2: Replace and Solve



Kick: replace that subgame by its expected utility
Go for 2: the subgame, replaced by its **mixed** NE payoffs

Michigan chooses: Go For 2

What Actually Happened in 2013



Photo: Shelby Lum, *The Lantern*

The Decision:
Hoke went for 2.
Consistent with SPNE!

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Did Hoke make a bad choice?

What Actually Happened in 2013



Photo: Shelby Lum, *The Lantern*

The Decision:
Hoke went for 2.
Consistent with SPNE!

The Play:
Michigan passed
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OSU wins 42–41

Did Hoke make a bad choice?
A 60% chance still
loses 40% of the time

The SPNE

The SPNE is fully specified by giving the probability of every action at **every** decision node:

Michigan: $\sigma_1^* = \left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ 0 & 1 & 3/5 & 2/5 \end{array} \right)$ OSU: $\sigma_2^* = \left(\begin{array}{cc} \text{SR} & \text{SP} \\ 1/2 & 1/2 \end{array} \right)$

The SPNE

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$$\text{Michigan: } \sigma_1^* = \left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ \underline{\quad} & \underline{\quad} & \underline{\quad} & \underline{\quad} \end{array}\right) \qquad \text{OSU: } \sigma_2^* = \left(\begin{array}{cc} \text{SR} & \text{SP} \\ \underline{\quad} & \underline{\quad} \end{array}\right)$$

Notice: we didn't write a probability for each **strategy**
We wrote one probability distribution per **information set**

The SPNE

The SPNE is fully specified by giving the probability of every action at **every** decision node:

$$\text{Michigan: } \sigma_1^* = \left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ 0 & 1 & 7/15 & 8/15 \end{array} \right) \qquad \text{OSU: } \sigma_2^* = \left(\begin{array}{cc} \text{SR} & \text{SP} \\ 1/2 & 1/2 \end{array} \right)$$

Notice: we didn't write a probability for each **strategy**
We wrote one probability distribution per **information set**

That's called a **behavioral strategy**

Selten's Theorem

Theorem (Selten, 1965)
Every finite extensive-form game with cardinal payoffs has **at least one** SPNE (possibly involving mixed strategies).

(**Finite game:** Finite # players, finite # of actions, tree doesn't go forever.)

Why?

- Every subgame is a finite game
- Nash: every finite game has a (mixed) NE
- So the SPNE algorithm can give a prediction at every subgame
- And the game is finite, so the algorithm doesn't run forever

Reinhard Selten shared the 1994 Nobel Prize with John Nash and John Harsanyi.

Behavioral Strategies

Michigan's Strategies

A strategy: one choice at every information set (Chapter 4)



Michigan has two information sets, two choices each:

$$S_{MI} = \{ \text{(Kick, Run)}, \text{(Kick, Pass)}, \text{(GoFor2, Run)}, \text{(GoFor2, Pass)} \}$$

- (Kick, Run) and (Kick, Pass) are different strategies, even though Kick precludes the Run/Pass decision.

Mixed strategy (Chapter 6): a probability for each of the 4 strategies

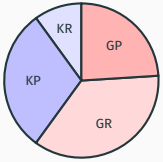
$$\begin{pmatrix} \text{(Kick, Run)} & \text{(Kick, Pass)} & \text{(GoFor2, Run)} & \text{(GoFor2, Pass)} \\ p & q & r & 1-p-q-r \end{pmatrix}$$

- Needs 3 numbers

Two Ways To Randomize

Mixed strategy

- Before the game: spin one big wheel over **complete plans**
- Then follow the plan



Behavioral strategy

- At each information set, when you get there: flip that set's coin
- One coin per information set



Behavioral strategy:

Kick	GoFor2	Run	Pass
p	$1-p$	q	$1-q$

Needs _____ numbers

Formal Definition: Behavioral Strategy

Definition (Bonanno 7.1.1)
A **behavioral strategy** for a player is a list of probability distributions, one for every information set of that player. These give the probability the player plays each action at each information set.

↖ every place where they can move

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Fact: If a player has only one information set then mixed = behavioral (like OSU)

$S_2 = \{SR, SP\}$ mixed strategy: $\begin{pmatrix} SR & SP \\ q & 1-q \end{pmatrix}$

behavioral strategy: $\begin{pmatrix} SR & SP \\ z & 1-z \end{pmatrix}$

Are They the Same?

Suppose Michigan's coach uses this mixed strategy:

$$\sigma_1 = \begin{pmatrix} \text{(Kick, Run)} & \text{(Kick, Pass)} & \text{(GoFor2, Run)} & \text{(GoFor2, Pass)} \\ 0.1 & 0.3 & 0.36 & 0.24 \end{pmatrix}$$

MIXED

0.4 0.6

Find a behavioral strategy that does the same thing:

BEHAVIORAL

$\sigma_1 = \left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ 0.4 & 0.6 & 0.6 & 0.4 \end{array}\right)$

Pr(Kick) = 0.4 Pr(GoFor2) = 0.6

Pr(Run | GoFor2) = $\frac{0.36}{0.6} = 0.6$ Pr(Pass | GoFor2) = $\frac{0.24}{0.6} = 0.4$

Notation: Pr(A | B) means "Probability of A, given *B* already occurred."
This is called a *conditional probability*.

Are They the Same?

Suppose Michigan's coach uses this mixed strategy:

$$\sigma_1 = \begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{GoFor2}, \text{Run}) & (\text{GoFor2}, \text{Pass}) \\ 0.1 & 0.3 & 0.36 & 0.24 \end{pmatrix}$$

Find a behavioral strategy that does the same thing:

$\Pr(\text{Kick}) =$ _____ $\Pr(\text{GoFor2}) =$ _____

$\Pr(\text{Run} \mid \text{GoFor2}) =$ _____ $\Pr(\text{Pass} \mid \text{GoFor2}) =$ _____

Check: $\Pr(\text{GoFor2 and Run}) =$ _____ $= 0.36$ ✓

Are They the Same?

Suppose Michigan's coach uses this mixed strategy:

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$\Pr(\text{Kick}) = _____$ $\Pr(\text{GoFor2}) = _____$

$\Pr(\text{Run} \mid \text{GoFor2}) = _____$ $\Pr(\text{Pass} \mid \text{GoFor2}) = _____$

Check: $\Pr(\text{GoFor2 and Run}) = _____ = 0.36$ ✓

Lesson: The 0.1 vs. 0.3 split never matters. The behavioral strategy doesn't require it.

Kuhn's Theorem

Theorem (Kuhn, 1953)
In games with **perfect recall**, for every mixed strategy there is a behavioral strategy that gives the same probability over terminal nodes

Recall perfect recall (Chapter 4):

- Never forget what you knew
- Never forget what you did

Takeaway: Use behavioral strategies. They're simpler, and nothing is lost

Notation: We'll use σ_i to describe either a mixed or behavioral strategy

Bonanno works through his own example (Figure 7.3) on pp. 239–241.

Two Ways To Be Unpredictable

Randomize **as you go** (behavioral):

- **Play-calling**: each down, run or pass?
- **Pitching**: each pitch, fastball or curveball?
- **Poker**: each betting round, bluff or fold?
- **Penalty kicks**: each kick, left or right?

Randomize **a whole plan in advance** (mixed):

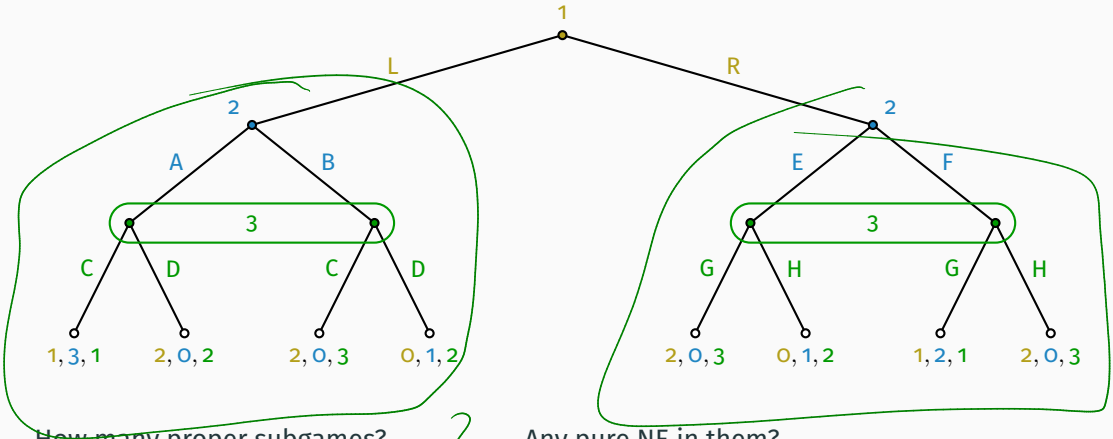
- **Security**: LAX's ARMOR software draws each day's checkpoint schedule
- **Inspections**: random drug-test lists, tax audits, restaurant inspections
- **Exams**: a random version of the test, handed out before it starts

Kuhn: with perfect recall, the two are equally powerful

A Bigger Example

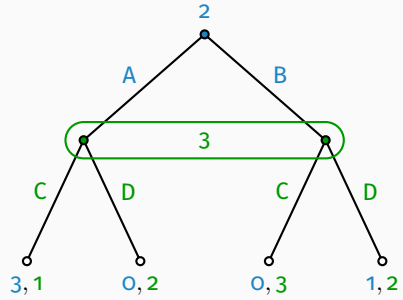
Figure 7.7

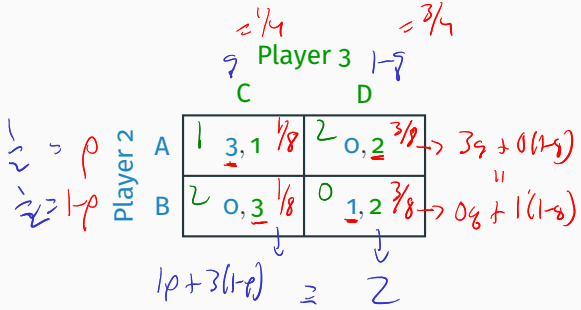
Find the SPNE



How many proper subgames? 2 Any pure NE in them?

Figure 7.8: The Left Subgame





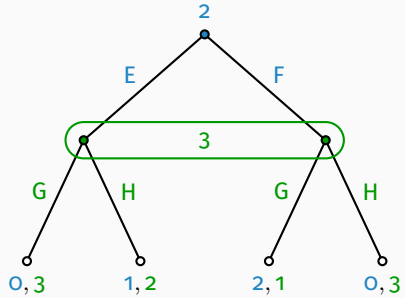
$q = \Pr(C)$ makes P2 indifferent:

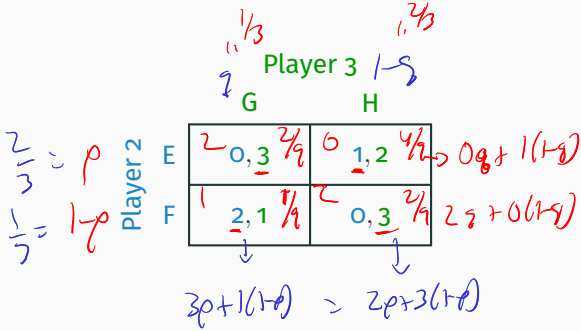
Payoffs (P1, P2, P3): (_____, _____, _____)

$p = \Pr(A)$ makes P3 indifferent:

P1's payoffs come from the tree

Figure 7.10: The Right Subgame



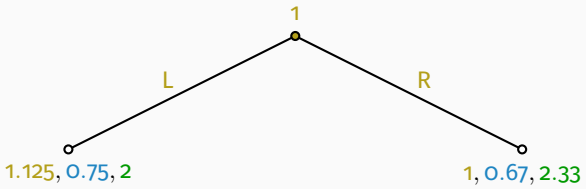


$q = \Pr(G)$ makes P2 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

$p = \Pr(E)$ makes P3 indifferent:

Figure 7.11: What's Left



Player 1 chooses: _____

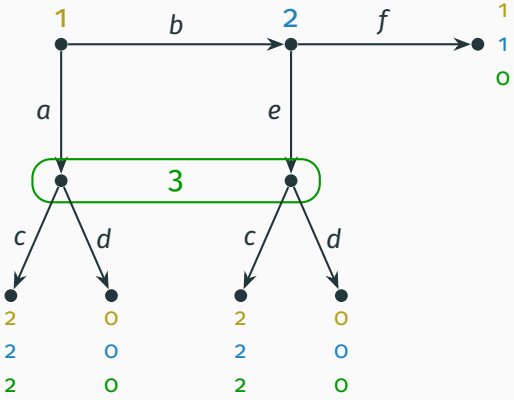
SPNE, in behavioral strategies:

$$\text{P1:} \begin{pmatrix} L & R \\ __ & __ \end{pmatrix} \qquad \text{P2:} \left(\begin{array}{cc|cc} A & B & E & F \\ __ & __ & __ & __ \end{array}\right)$$

$$\text{P3:} \left(\begin{array}{cc|cc} C & D & G & H \\ __ & __ & __ & __ \end{array}\right)$$

A Critique of SPNE
Due to Selten Himself

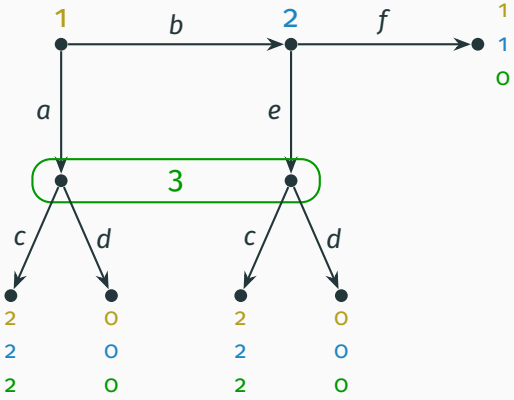
A Critique of SPNE: Selten's Horse



Proper subgames? _____

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

A Critique of SPNE: Selten's Horse

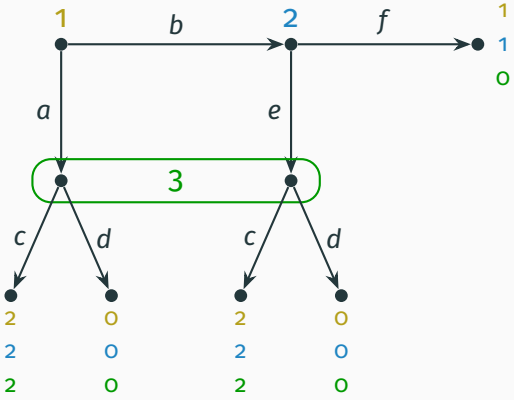


Proper subgames? _____

So SPNE = NE

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A Critique of SPNE: Selten's Horse



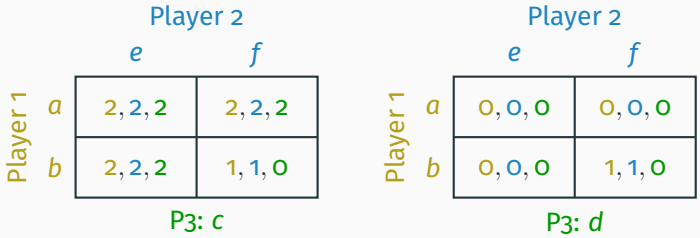
Proper subgames? _____

So SPNE = NE

To find the NE: write the strategic form

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

Selten's Horse: Strategic Form



P1 picks the row, P2 the column, P3 the matrix

Selten's Horse: Strategic Form

P3: *c*

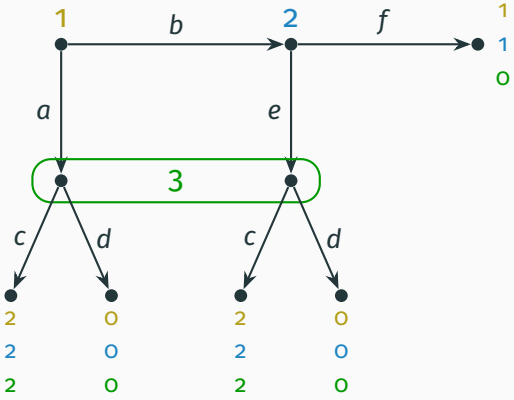
Player 1 \ Player 2	*e*	*f*
a	2, 2, 2	2, 2, 2
b	2, 2, 2	1, 1, 0

P3: *d*

Player 1 \ Player 2	*e*	*f*
a	0, 0, 0	0, 0, 0
b	0, 0, 0	1, 1, 0

Pure NE: _____ _____ _____ _____

Selten's Horse



Pure NE (all are SPNE):

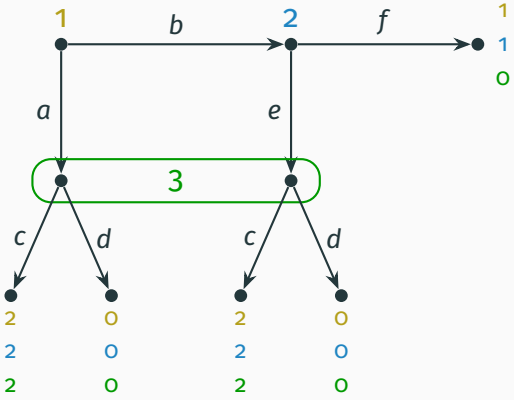
(a, e, c) (a, f, c)

(b, e, c) (b, f, d)

Are they all reasonable???

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

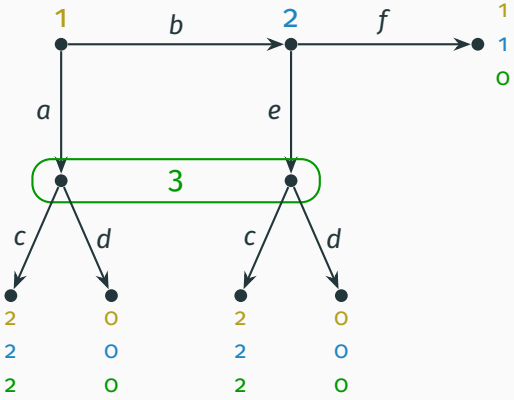
Problem 1: (a, f, c)



P1 plays a. P2 never moves

So P2's plan f costs P2 nothing

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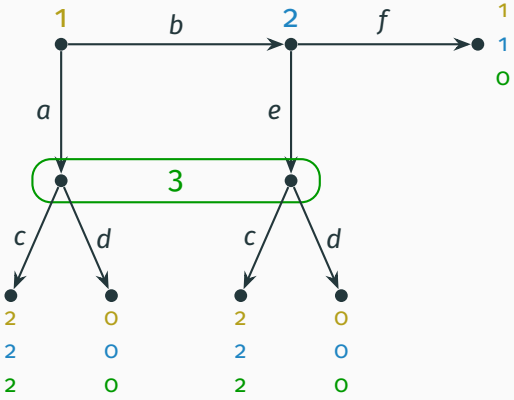
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So P2's plan f costs P2 nothing

But **if** P2 had to move?

e gives _____ f gives _____

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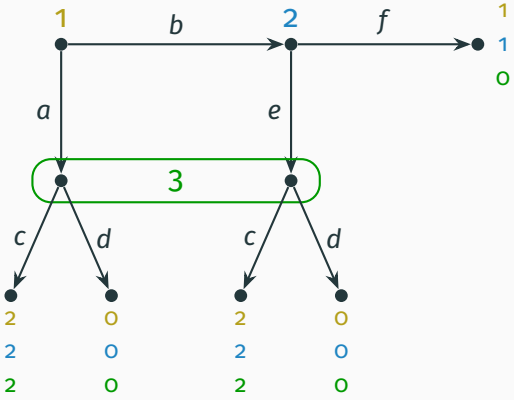
So P2's plan f costs P2 nothing

But **if** P2 had to move?

e gives _____ f gives _____

f is a bad plan. SPNE doesn't notice

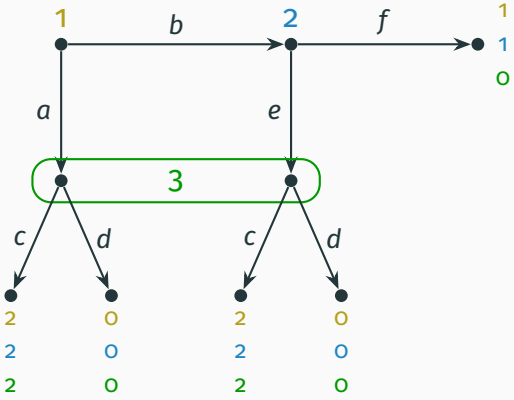
Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

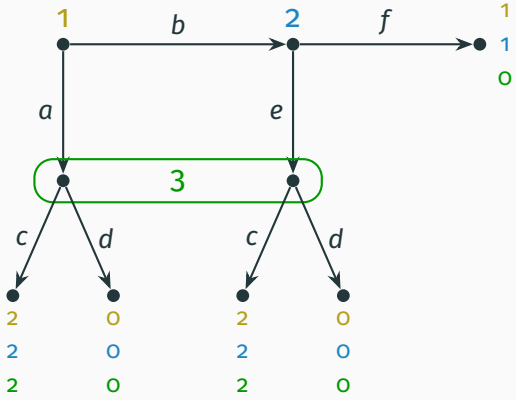
But **if** P3 had to move?

Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

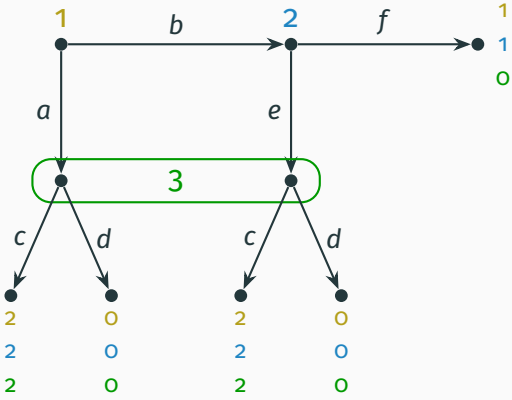
Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

Lesson: d is dominated as a **choice**, not as a **strategy**
Unreached infosets escape the test

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

So What's Missing?

SPNE checks for rationality only in **subgames**

- A subgame can't cut an information set
- So choices at big infosets are only checked if they're on the path of play

The fix: Every choice should be sensible, **even off the path of play**

- P3 doesn't know which node she's at
  $\Rightarrow$ she needs a **belief** about where she is

Thus: $\{\text{better equilibria}\} \subseteq SPNE \subseteq NE$

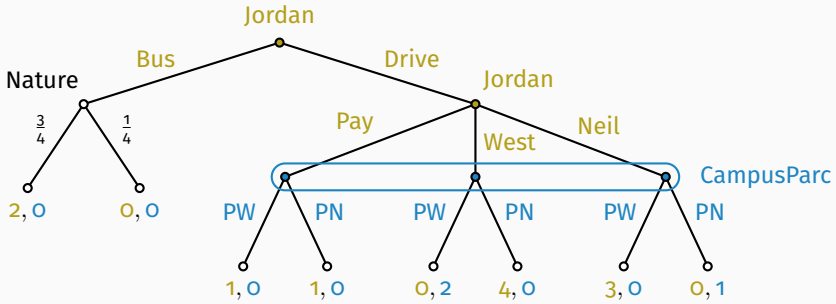
Bonanno Part IV (Chapters 11–13): weak sequential, sequential, and perfect Bayesian equilibrium.

Let’s Work One Out

Gameday Parking, Part 2

Jordan first decides how to get to the game:

- **Bus** (COTA): on time with chance $\frac{3}{4}$ (payoff 2), else late (payoff 0)
- **Drive**: then Chapter 6's parking game. CampusParc can't see where Jordan parked



Gameday Parking, Part 2

1. List Jordan's pure strategies. How many numbers for a mixed strategy? A behavioral one?

2. What is Jordan's expected payoff from Bus?

3. Solve the parking subgame (recall Chapter 6). Payoffs?

4. Find the SPNE. Write it in behavioral strategies

5. How reliable must the bus be before Jordan takes it?

What Have We Learned, Charlie Brown?

- Nature in a tree? Replace the lottery with its **expected utility**
- **Behavioral strategy**: one probability distribution per information set
  - Mixed: spin one wheel over whole plans. Behavioral: flip as you go
- **Kuhn (1953)**: with perfect recall, behavioral $=$ mixed
- **Selten (1965)**: with cardinal payoffs, a SPNE always exists
  - Solve subgames with mixed NE when there's no pure one
- SPNE can still allow silly choices at unreached infosets (Selten's horse)

Reading & Practice

- Bonanno, *Game Theory*, Chapter 7
  - §7.1: read for behavioral strategies and Kuhn's theorem. Skim the formal "game with cardinal payoffs" part
  - §7.2–7.3: read carefully
- Exercises: §7.4.1 (7.1, 7.2, 7.3), §7.4.2 (7.5, 7.7)
  - 7.4 and 7.6 review Chapter 5's utility calibration
  - 7.8 is harder. 7.9 (the mugger) is a challenge
- Solutions in §7.5. Try them first